# 100mA, 45V Schottky Barrier Diode

## FEATURES

- AEC-Q101 qualified
- Low Capacitance
- Low forward voltage drop
- Surface mound device type
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free

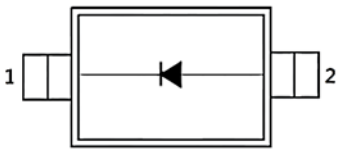
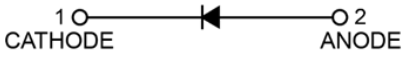
| KEY PARAMETERS | | |
|---|---|---|
| **PARAMETER** | **VALUE** | **UNIT** |
| $I_F$ | 100 | mA |
| $V_{RRM}$ | 45 | V |
| $I_{FSM}$ | 1 | A |
| $T_{J\ MAX}$ | 125 | °C |
| $V_F$ at $I_F$=10mA | 0.45 | V |
| Configuration | Single die | |

## APPLICATIONS

- Adapters
- For switching power supply
- Low stored charge
- Inverter

## MECHANICAL DATA

- Case: SOD-323
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: Indicated by cathode band
- Weight: 4.5mg (approximately)

| PACKAGE: SOD-323 | PIN CONFIGURATION | CIRCUIT DIAGRAM |
|---|---|---|
|  |  |  |



## ABSOLUTE MAXIMUM RATINGS ($T_A$ = 25°C unless otherwise noted)

| PARAMETER | | SYMBOL | VALUE | UNIT |
|---|---|---|---|---|
| Power dissipation(1) | | $P_D$ | 200 | mW |
| Peak reverse voltage | | $V_{RM}$ | 45 | V |
| Reverse voltage | | $V_R$ | 40 | V |
| Forward current | | $I_F$ | 100 | mA |
| Non-repetitive peak forward surge current | t = 8.3ms | $I_{FSM}$ | 1 | A |
| Junction temperature | | $T_J$ | -55 to +125 | °C |
| Storage temperature | | $T_{STG}$ | -55 to +150 | °C |

**Note:**

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

| THERMAL PERFORMANCE | | | |
|---|---|---|---|
| **PARAMETER** | **SYMBOL** | **TYP** | **UNIT** |
| Junction-to-ambient thermal resistance(1) | $R_{\Theta JA}$ | 500 | °C/W |

**Thermal Performance Note:**

1. Device mounted on an FR4 Printed-Circuit Board (PCB), single-sided copper, tin-plated and standard footprint

| ELECTRICAL SPECIFICATIONS ($T_A$ = 25°C unless otherwise noted) | | | | | | |
|---|---|---|---|---|---|---|
| **PARAMETER** | **CONDITIONS** | **SYMBOL** | **MIN** | **TYP** | **MAX** | **UNIT** |
| Forward voltage(1) | $I_F$ = 10mA | $V_F$ | - | - | 0.45 | V |
| Reverse breakdown voltage | $I_R$ = 100μA | $V_{BR}$ | 45 | - | - | V |
| Reverse current(2) | $V_R$ = 10V | $I_R$ | - | - | 1 | μA |
| Junction capacitance | f = 1MHz, $V_R$ = 10V | $C_J$ | - | 6 | - | pF |

**Notes:**

1. Pulse test with PW=0.3 ms
2. Pulse test with PW=30 ms

| ORDERING INFORMATION | | |
|---|---|---|
| **ORDERING CODE** | **PACKAGE** | **PACKING** |
| RB500V-40H RRG | SOD-323 | 3,000 / 7” Tape & Reel |

## CHARACTERISTICS CURVES

($T_A$ = 25°C unless otherwise noted)

**Fig.1 Power Dissipation Curve**

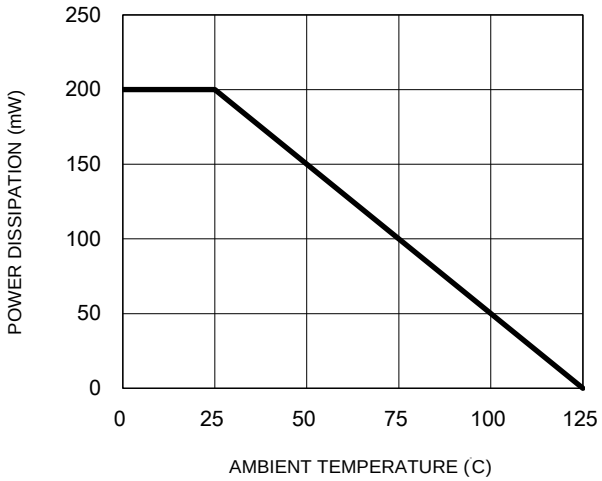



**Fig.2 Typical Junction Capacitance**

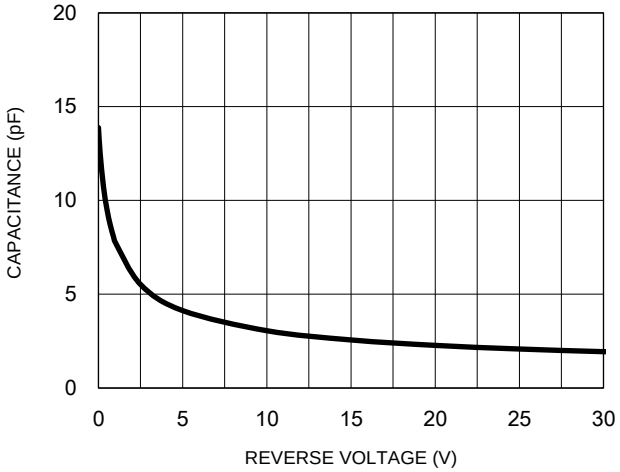



**Fig.3 Typical Reverse Characteristics**

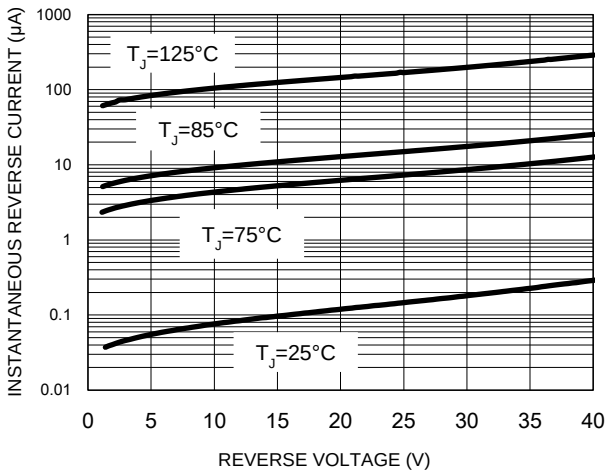
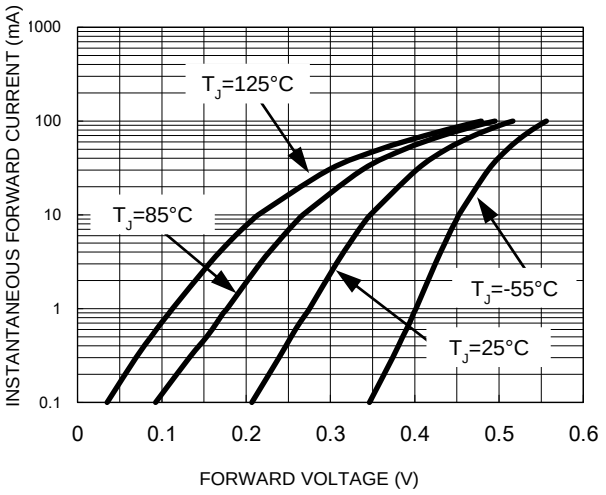
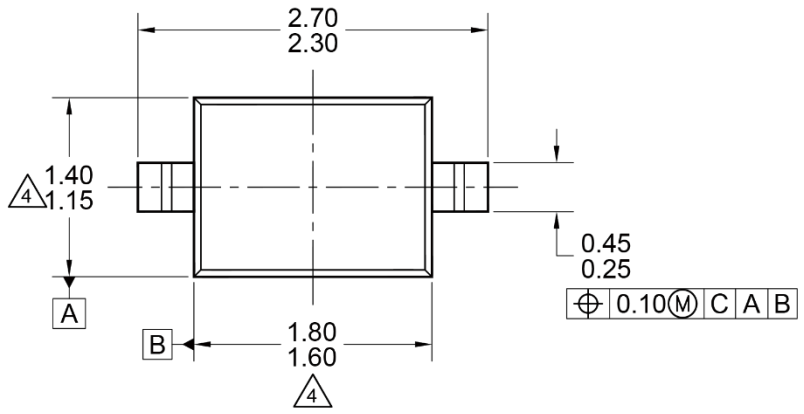


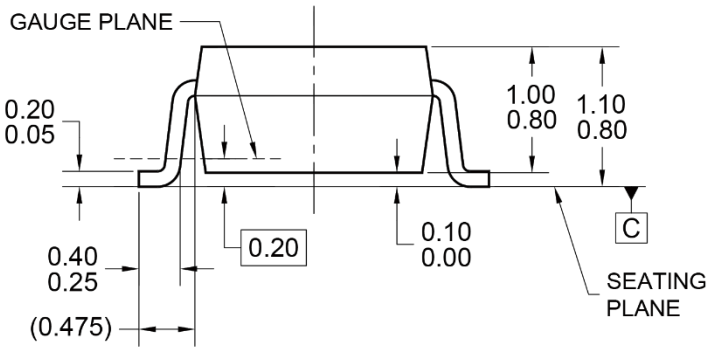

**Fig.4 Typical Forward Characteristics**

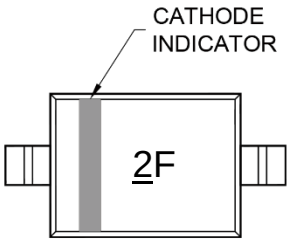


## PACKAGE OUTLINE DIMENSION

**SOD-323**

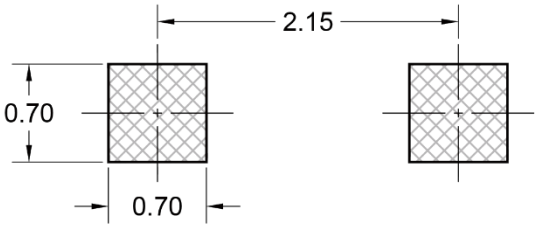



MARKING DIAGRAM

2 = Device marking
F = Factory code




SUGGESTED PAD LAYOUT

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. PACKAGE OUTLINE REFERENCE: EIAJ ED-7500A, SC-76A.
4. MOLDED PLASTIC BODY LATERAL DIMENSIONS DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS.
5. DWG NO. REF: HQ2SD07-SOD323-113 REV A.

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